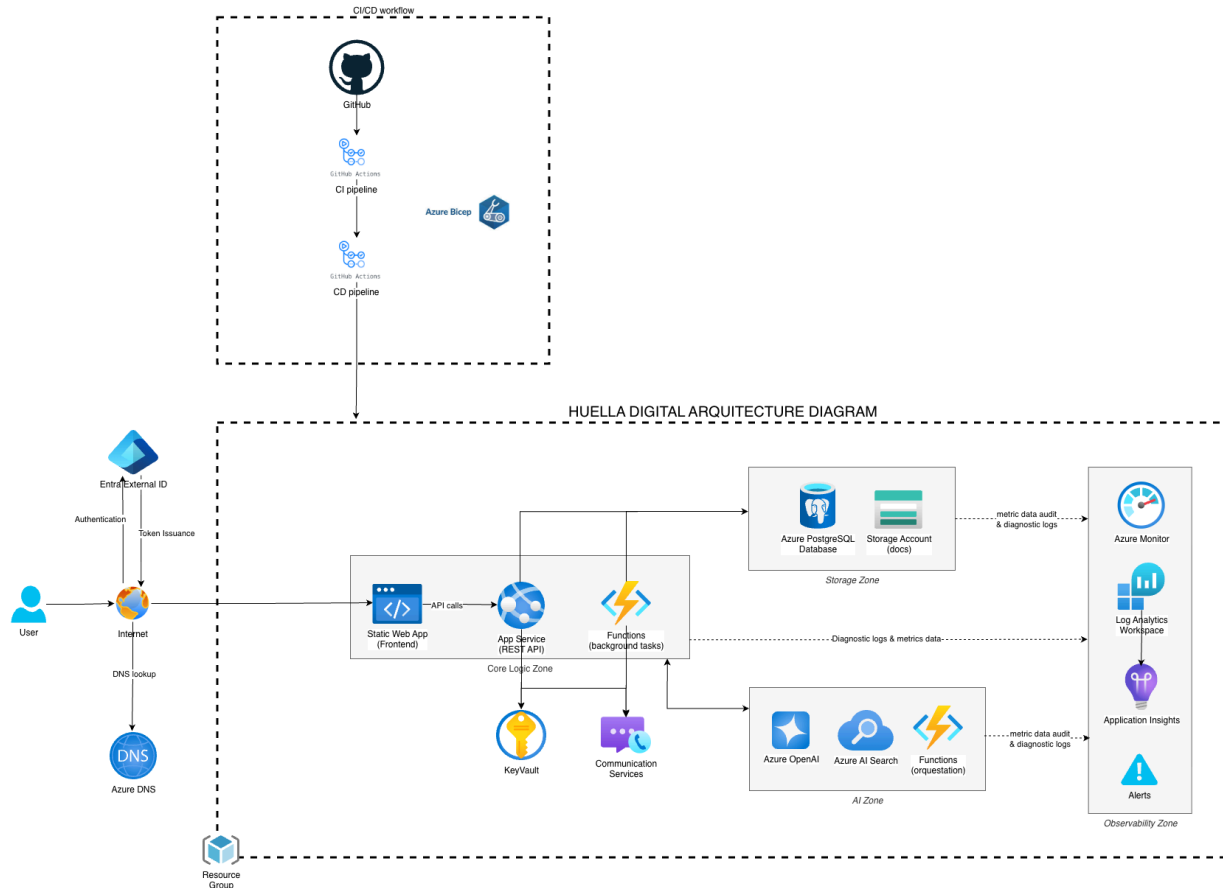


HuellaLatam – Infrastructure Provisioning & Deployment Model

This document describes the **infrastructure provisioning and deployment model** for the Huella Latam platform. It defines **how cloud resources are created, managed, and deployed**, emphasizing an **Infrastructure as Code (IaC)** approach using **Azure Bicep** and **automated CI/CD pipelines**.

The objective is to ensure **consistency, security, auditability, and repeatability** across environments, while clearly outlining the responsibilities and prerequisites required to enable automated deployments.

Architecture Diagram



Delivered Documentation Package

The following files are shared with the IT team for **review, validation, and alignment**:

- Cloud Infrastructure Assessment, a folder which contains:
 - App Usage Assumptions & Estimations
 - Azure Service Assessments (per service)
- Consolidated Azure Service Requirements
- Azure Pricing Calculator Output (Excel file)

These documents define **what infrastructure is required** and **how it is expected to behave**.

Infrastructure Provisioning – Mandatory Approach

All infrastructure for this project must be provisioned using Infrastructure as Code (IaC).

Manual resource creation is explicitly out of scope

- Resources must **not** be created manually in the Azure Portal
- Resources must **not** be pre-created and handed over
- Any manually created infrastructure will be considered **non-compliant**

Manual creation would:

- Break environment consistency
- Introduce configuration drift
- Make environments non-reproducible
- Increase operational and security risk
- Prevent reliable promotion from STAGING to PROD

For these reasons, **manual provisioning is explicitly out of scope** for this project.

Infrastructure as Code (IaC) Standard

- Tooling: **Azure Bicep**
- Execution: **Automated CI/CD pipelines**
- Source of truth: **Git repository**
- Environments: **Reproducible, auditable, version-controlled**

All changes to infrastructure are performed:

- Through code
- Through pull requests
- Through controlled deployments

Infrastructure as Code (Bicep) allows us to:

- Provision **identical, reproducible environments**
- Version-control all infrastructure changes
- Enforce security, naming, tagging, and configuration standards
- Audit infrastructure changes over time
- Recreate environments safely if needed
- Reduce human error and misconfiguration
- Align with Azure and industry best practices

In short: **the documents describe *what* is needed; Bicep defines *how* it is created.**

CI/CD Pipelines

- **GitHub** is the source control system
 - **GitHub Actions** is used for:
 - Infrastructure deployment (Bicep)
 - Application deployments
 - Pipelines handle:
 - the creation and update of the STAGING environments
 - Repeatable and idempotent deployments
-

Prerequisites Required From IT Team

The items listed below are prerequisites that **must be completed by the IT team** before infrastructure deployment can begin.

These are not informational; they require concrete actions (subscription setup, access grants, identity configuration) to enable automated provisioning via Infrastructure as Code.

Until these prerequisites are fulfilled, CI/CD pipelines and Bicep deployments cannot be executed.

1 Azure Subscription

- Existing Azure subscription
 - Permission to deploy resources programmatically
 - Subscription policies (if any) shared in advance
-

2 Access via Entra ID + PIM

- Access granted through **Microsoft Entra ID**
 - **Privileged Identity Management (PIM)** preferred
 - Required roles (scope: subscription or RG):
 - Contributor (minimum)
 - User Access Administrator (if required by policy)
-

3 Resource Group Management

One of the following:

- Permission to **create resource groups via IaC**
- Or ownership of **empty, pre-created RGs**

In all cases:

- RG contents are managed exclusively by Bicep
-

4 Entra External ID / Authentication Setup

- Entra tenant available
- Initial setup of:
 - App registrations
 - Required claims exposure
- Ability for the project team to:

- Configure claims
 - Integrate authentication flows
-

Azure Resource Provider Availability

The subscription must allow registration and usage of the following resource providers (as applicable):

- Microsoft.Web
- Microsoft.Storage
- Microsoft.DBforPostgreSQL
- Microsoft.Search
- Microsoft.CognitiveServices
- Microsoft.Insights
- Microsoft.KeyVault
- Microsoft.Communication
- Microsoft.Operationallnsights

These are required for Bicep deployments to succeed.

OpenID Connect (OIDC) Federation between GitHub and Azure

GitHub Actions authenticates to Azure using **OIDC (federated identity)**.

- No secrets stored in GitHub
- No client secrets or passwords
- Short-lived tokens issued at runtime
- Full auditability and compliance

This setup is **mandatory**.

Responsibilities Split

Client IT Team

- Provide subscription, access, identity foundations
- Review and approve architecture & requirements

Project Team

- Provision infrastructure via Bicep
 - Deploy applications via CI/CD
 - Maintain infrastructure definitions as code
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Final Note

The attached documents and diagrams describe **the required target state**.

The **only supported path** to reach that target state is: Infrastructure as Code + GitHub Actions + Azure Bicep

Any deviation from this approach introduces risk and is not aligned with the project's delivery model.